



CUBAN NATIONAL CHEMISTRY CONTEST

“To work in the sterile when one can do the useful; to deal with the easy when one has spirit to try the difficult, is to strip the dignity from the talent.”

José Martí

Name(s): _____ Last name(s): _____

School: _____

Municipality: _____ Province: _____

CODE: _____ Test: **DAY 1. COMMON EXAM**

Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 1 problem set, with a common exam for the three grades. The exam has 3 problems (problem 1, 30 items; problem 2, 4 items; problem 3, 2 items) and a total of 5 pages.
- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem its scoring (from a total of 100 points), each item scoring (in marks) is indicated.

Success!



Ministerio de Educación
República de Cuba



CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's Constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gasses' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	Thermodynamics First Principle	$Q = \Delta E + W$
Cero in the Celsius scale	273,15 K	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Water ionic product at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$	$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF THE ELEMENTS

																	13 3A		14 4A	15 5A	16 6A	17 7A	18 8A								
1 1A 1 H 1.008																					2 He 4.003										
3 Li 6.941	4 Be 9.012												5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18													
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95														
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80														
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3														
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)														
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (269)	111 Rg (272)	112 Uub (277)		114 Uuq (277)		116 Uuh (277)		118 Uuo (277)														
																		58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
																		90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Problem 1. Miscellaneous (30 points)

1.1	...	Total	Total
1	idem	30	30
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1.1. Which of the following substances dissolves better in apolar solvents?

- A) glucose B) graphite C) lithium fluoride D) octasulfur

1.2. 0,250 g of element X react with excess fluorine to produce 0,547 g of fluoride XF_6 . ¿Which is the element X?

- A) Cr B) Mo C) S D) Te

1.3. How many moles of Na^+ ions are there in 20 mL of a Na_3PO_4 0,40 $\text{mol}\cdot\text{L}^{-1}$ solution?

- A) 0,0080 B) 0,024 C) 0,050 D) 0,20

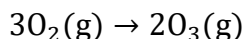
1.4. Which solid reacts with diluted hydrochloric acid at 25 °C to form a gas denser than air?

- A) Zn B) $\text{Pb}(\text{NO}_3)_2$ C) NaBr D) NaHCO_3

1.5. How many molecules are in the 2,5 g of O_2 that an average person inhales every minutes?

- A) $1,9\cdot 10^{22}$ B) $3,8\cdot 10^{22}$ C) $4,7\cdot 10^{22}$ D) $9,4\cdot 10^{22}$

1.6. The rate of formation of O_3 is $2,0\cdot 10^{-7} \text{ mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$ for the reaction



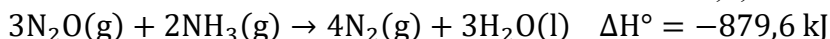
Which is the rate of consumption of O_2 in units $\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$?

- A) $1,3\cdot 10^{-7}$ B) $2,0\cdot 10^{-7}$ C) $3,0\cdot 10^{-7}$ D) $4,5\cdot 10^{-7}$

1.7. The molar mass of a gas with a density of 5,8 $\text{g}\cdot\text{L}^{-1}$ at 25 °C and 100 kPa is about

- A) 10 $\text{g}\cdot\text{mol}^{-1}$ B) 20 $\text{g}\cdot\text{mol}^{-1}$ C) 150 $\text{g}\cdot\text{mol}^{-1}$ D) 190 $\text{g}\cdot\text{mol}^{-1}$

1.8. Which is the ΔH_f° of N_2O in $\text{kJ}\cdot\text{mol}^{-1}$? $\Delta H_f^\circ/\text{kJ}\cdot\text{mol}^{-1}$: $\text{NH}_3 = -45,9$; $\text{H}_2\text{O} = -241,8$.



- A) +246 B) +82 C) -82 D) -246

1.9. When isoelectronic ions K^+ , Ca^{2+} y Cl^- are ordered in consecutive increasing radii, the correct order is

- A) K^+ , Ca^{2+} , Cl^- B) K^+ , Cl^- , Ca^{2+} C) Cl^- , Ca^{2+} , K^+ D) Ca^{2+} , K^+ , Cl^-

1.10. Which element from group 2 possesses distinct chemical properties from the others?

- A) Be B) Ca C) Sr D) Ba

1.11. Which atom has the highest ionization energy?

- A) Na B) K C) Mg D) Ca

1.12. All the following species have the same number of valence electrons as NO_3^- , except for

- A) CO_3^{2-} B) HCO_3^- C) NF_3 D) SO_3

1.13. Which list does not contain any ionic species?

- A) NH_4Cl , OF_2 , H_2S B) CO_2 , Cl_2 , CCl_4 C) BF_3 , AlF_3 , TlF_3 D) I_2 , CaO , CH_3Cl

1.14. How many carbon-carbon bonds are in a molecule of 2-methyl-2-butanol?

- A) 2 B) 3 C) 4 D) 5

1.15. When the following species are ordered according to increasing bond energy, which is the correct sequence?

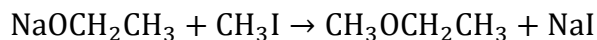
- A) N_2 , O_2 , F_2 B) F_2 , O_2 , N_2 C) O_2 , F_2 , N_2 D) O_2 , N_2 , F_2

- 1.16.** The ionic specie PCl_4^- has a molecular geometry
A) tetrahedral B) irregular tetrahedral C) squared D) trigonal bipyramidal
- 1.17.** Which of the following molecules is not polar? 1. NCl_3 ; 2. SO_3 ; 3. PCl_5 .
A) only 1 B) only 2 C) only 1 and 3 D) only 2 and 3
- 1.18.** Which are the hybridizations of the highlighted atoms in the hydrocarbon? $\underline{\text{C}}\text{H}_3\underline{\text{C}}\text{HCH}_2$
A) sp^3 , sp B) sp^3 , sp^2 C) sp^2 , sp^2 D) sp , sp^2
- 1.19.** Which of the following ions are colored in an aqueous solution? 1. Fe^{3+} ; 2. Ni^{2+} ; 3. Al^{3+} .
A) only 1 B) only 3 C) only 1 and 2 D) 1, 2 and 3
- 1.20.** Which substance is stored under water to prevent contact and reaction with air?
A) bromine B) lithium C) mercury D) phosphorous
- 1.21.** *t*-butylmethylether, $\text{C}_5\text{H}_{12}\text{O}$, is added to gas to promote clean combustion. How much oxygen is required for the complete combustion of 1,0 mol of this compound?
A) 4,5 mol B) 6,0 mol C) 7,5 mol D) 8,0 mol
- 1.22.** Which gas is produced when diluted aqueous nitric acid reacts with silver?
A) NO B) H_2 C) NH_3 D) N_2
- 1.23.** Which kind of solid can be associated with relatively low melting points and electric conductivity, in general?
A) ionic B) metallic C) molecular D) atomic covalent
- 1.24.** Use bond energies to estimate the ΔH° for the reaction $\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}_2(\text{g})$.
 $\varepsilon(\text{A}-\text{B})/\text{kJ}\cdot\text{mol}^{-1}$: $\text{H}-\text{H} = 436$; $\text{O}-\text{O} = 142$; $\text{O}=\text{O} = 499$; $\text{H}-\text{O} = 460$.
A) -127 B) -209 C) -484 D) -841
- 1.25.** What happens when metallic zinc is added to a solution containing magnesium nitrate and silver nitrate? 1. Zn is oxidized; 2. Mg^{2+} is reduced; 3. Ag^+ is reduced; 4. No reaction occurs.
A) only 1 and 2 B) only 1 and 3 C) only 1, 2 and 3 D) only 4
- 1.26.** How many unpaired electrons has an isolated Fe^{2+} ion in the ground state?
A) 0 B) 2 C) 4 D) 6
- 1.27.** What is the formal charge in the central atom of N_2O ?
A) -1 B) 0 C) $+1$ D) $+2$
- 1.28.** How many isomers correspond to the molecular formula $\text{C}_3\text{H}_8\text{O}$?
A) 2 B) 3 C) 4 D) 5
- 1.29.** When the species NH_3 , NH_2^- and NH_4^+ are ordered with increasing H-N-H bond angle, the right order is
A) NH_3 , NH_2^- , NH_4^+ B) NH_4^+ , NH_2^- , NH_3 C) NH_3 , NH_4^+ , NH_2^- D) NH_2^- , NH_3 , NH_4^+
- 1.30.** The molar mass of the air (71% of N_2 and 29% of O_2 , volume/volume) in $\text{g}\cdot\text{mol}^{-1}$ is
A) 28 B) 31 C) 29 D) 32

Problem 2. Kinetics (10 points)

2.1	2.2	1.3	1.4	Total	Total
6	6	4	4	20	10

The following table shows the results of a kinetic study of the reaction between sodium ethoxide and methyl iodide in ethanol at 298 K. It is known that the kinetics of this reaction has an Arrhenius-type behavior.



Experiment	$c_0(\text{NaOEt})/\text{mol}\cdot\text{L}^{-1}$	$c_0(\text{MeI})/\text{mol}\cdot\text{L}^{-1}$	$v_0/\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$
1	1,0	2,0	0,00245
2	2,5	0,5	0,00153
3	2,0	1,5	0,00366
4	3,0	2,0	0,00734

2.1. Calculate the partial reaction order with respect to sodium ethoxide and methyl iodide.

2.2. Calculate the rate constant at 298 K.

If you cannot solve the previous item, use the value $k = 5,0 \cdot 10^{-3} \text{ u}$ for the rate constant when solving the rest of the problem.

2.3. Calculate the activation energy E_A if the reaction rate increases 4,8-fold when the temperature rises to 313 K.

If you cannot solve the previous item, use the value $E_A = 1,0 \cdot 10^5 \text{ J}\cdot\text{mol}^{-1}$ for the activation energy when solving the rest of the problem.

2.4. Calculate the pre-exponential factor A.

Problem 3. Stoichiometry (10 points)

3.1	3.2	Total	Total
5	15	20	10

4,00 g of a mixture containing octasulfur, zinc sulfide and iron(II) disulfide (FeS_2) were reacted with enough oxygen. As a result, all the sulfur contained in the mixture was converted into 1,90 L of sulfur dioxide, measured at 25 °C and 101,325 kPa. The resultant solid residual was dissolved in sulfuric acid. After concentrating and heating to a high temperature, 4,66 g of a mixture of zinc sulfate and iron(III) sulfate was obtained.

3.1. Write the equations corresponding to all the chemical reactions described.

3.2. Calculate the composition (in mass percent) of the initial mixture.

$\text{M}\cdot\text{g}\cdot\text{mol}^{-1}$: $\text{S}_8 = 256,56$; $\text{ZnS} = 97,48$; $\text{FeS}_2 = 119,99$; $\text{ZnSO}_4 = 161,48$; $\text{Fe}_2(\text{SO}_4)_3 = 399,91$.